Natural Language Processing

ACTL3143 & ACTL5111 Deep Learning for Actuaries

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Lecture Outline

- **Natural Language Processing**
- How Computers View Text
- Car Crash Police Reports
- Text Vectorisation with Bag of Words
- Discard Infrequent Words and Stop Words
- Merge Together Similar Words
- Use a Continuous Representation
- Word Embeddings
- Demo of Word Embeddings
- Recap

What is NLP?

A field of research at the intersection of computer science, linguistics, and artificial intelligence that takes the *naturally spoken or written language* of humans and *processes it with machines* to automate or help in certain tasks.

Applications of NLP in Industry

1) Classifying documents: Using the language within a body of text to classify it into a particular category, e.g.:

- Grouping emails into high and low urgency
- Movie reviews into positive and negative sentiment (i.e. *sentiment analysis*)
- Company news into bullish (positive) and bearish (negative) statements

2) Machine translation: Assisting language translators with machine-generated suggestions from a source language (e.g. English) to a target language

3) Search engine functions, including:

- Autocomplete
- Predicting what information or website user is seeking

4) Speech recognition: Interpreting voice commands to provide information or take action. Used in virtual assistants such as Alexa, Siri, and Cortana

Deep learning & NLP?

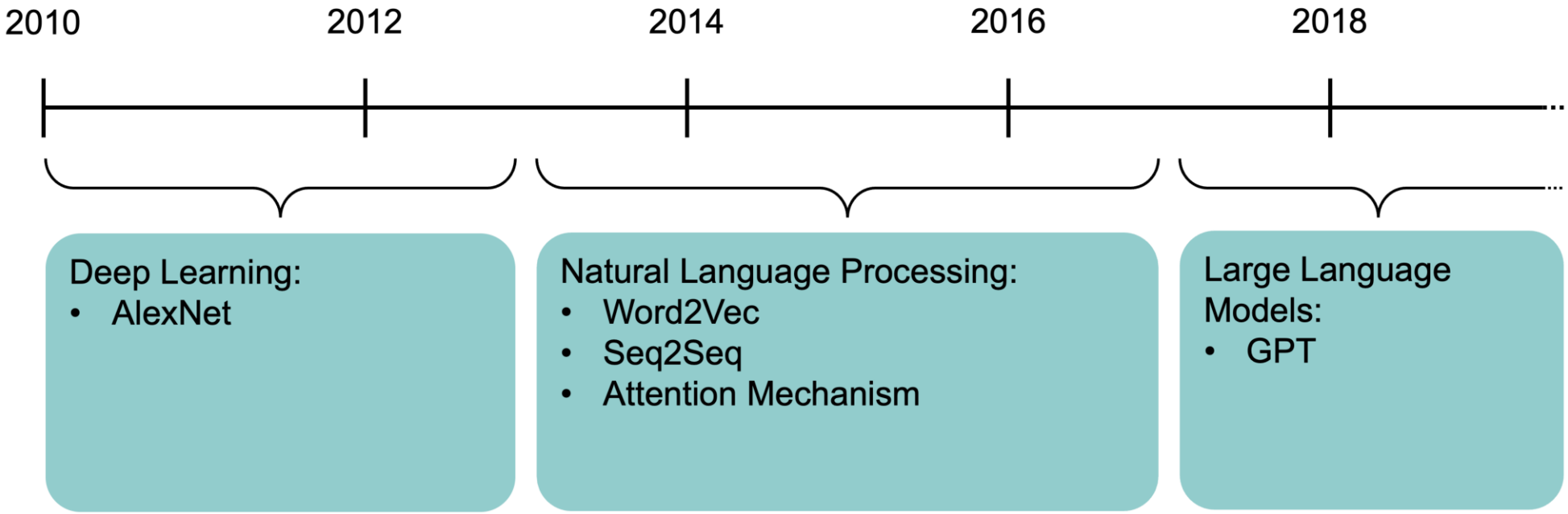
Simple NLP applications such as spell checkers and synonym suggesters *do not require deep learning* and can be solved with *deterministic, rules-based code* with a dictionary/thesaurus.

More complex NLP applications such as classifying documents, search engine word prediction, and chatbots are complex enough to be solved using deep learning methods.

NLP in 1966–1973

> "A typical story occurred in early machine translation efforts, which were generously funded by the U.S. National Research Council in an attempt to speed up the translation of Russian scientific papers in the wake of the Sputnik launch in 1957. It was thought initially that simple syntactic transformations, based on the grammars of Russian and English, and word replacement from an electronic dictionary, would suffice to preserve the exact meanings of sentences. The fact is that accurate translation requires background knowledge in order to resolve ambiguity and establish the content of the sentence. The famous retranslation of *"the spirit is willing but the flesh is weak"* as *"the vodka is good but the meat is rotten"* illustrates the difficulties encountered. In 1966, a report by an advisory committee found that "there has been no machine translation of general scientific text, and none is in immediate prospect." All U.S. government funding for academic translation projects was canceled."
>
> — Russell & Norvig (2021, p. 21)

High-level history of deep learning



A brief history of deep learning.

The attention mechanism (Bahdanau et al., 2015) and the Transformer architecture (Vaswani et al., 2017) underpin modern large language models.

Source: Melissa Renard (2025)

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Python imports

```
import random
from pathlib import Path

import zipfile
import ast, dis, io, tokenize
from urllib.request import urlretrieve

import numpy as np
import numpy.random as rnd
import pandas as pd
import spacy

from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder

import keras
from keras.callbacks import EarlyStopping
from keras.layers import Dense, Input
from keras.metrics import SparseTopKCategoricalAccuracy
from keras.models import Sequential
```

Also need to run `python -m spacy download en_core_web_trf`.

How the computer sees text

Spot the odd one out:

```
[112, 97, 116, 114, 105, 99, 107, 32, 108, 97, 117, 98]
[80, 65, 84, 82, 73, 67, 75, 32, 76, 65, 85, 66]
[76, 101, 118, 105, 32, 65, 99, 107, 101, 114, 109, 97, 110]
```

Generated by:

```python
print([ord(x) for x in "patrick laub"])
print([ord(x) for x in "PATRICK LAUB"])
print([ord(x) for x in "Levi Ackerman"])
```

The `ord` built-in turns characters into their ASCII form.

Question

The largest value for a character is 127, can you guess why?

American Standard Code for Information Interchange

Dec	Char	Esc
0	[Null]	\0
1	[Start of Heading]	
2	[Start of Text]	
3	[End of Text]	
4	[End of Transmission]	
5	[Enquiry]	
6	[Acknowledge]	
7	[Bell]	\a
8	[Backspace]	\b
9	[Horizontal Tab]	\t
10	[Line Feed]	\n
11	[Vertical Tab]	\v
12	[Form Feed]	\f
13	[Carriage Return]	\r
14	[Shift Out]	
15	[Shift In]	

Dec	Char
32	[Space]
33	!
34	”
35	#
36	$
37	%
38	&
39	’
40	(
41	)
42	*
43	+
44	,
45	-
46	.
47	/

Dec	Char
64	@
65	A
66	B
67	C
68	D
69	E
70	F
71	G
72	H
73	I
74	J
75	K
76	L
77	M
78	N
79	O

Dec	Char
96	`
97	a
98	b
99	c
100	d
101	e
102	f
103	g
104	h
105	i
106	j
107	k
108	l
109	m
110	n
111	o

“I knew she’d have an ASCII table in there somewhere. All computer geeks do.” (The Martian)

Random strings

The built-in `chr` function turns numbers into characters.

```
1 rnd.seed(1)
```

```
1 chars = [chr(rnd.randint(32, 127)) for _ in range(10)]
2 chars
```

```
['E', ',', 'h', ')', 'k', '%', 'o', '`', '0', '!']
```

```
1 " ".join(chars)
```

```
'E , h ) k % o ` 0 !'
```

```
1 "".join([chr(rnd.randint(32, 127)) for _ in range(50)])
```

```
"lg&9R42t+⩽.Rdww~v-)'_]6Y! \\q(x-Oh>g#f5QY#d8Kl:TpI"
```

```
1 "".join([chr(rnd.randint(0, 128)) for _ in range(50)])
```

```
'R\x0f@D\x19obW\x07\x1a\x19h\x16\tCg~\x17}d\x1b%9S&\x08 "\n\x17\x0foW\x19Gs\\J>.
X\x177AqM\x03\x00x'
```

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Escape characters

```
1 print("Hello,\tworld!")
```

```
Hello,	world!
```

```
1 print("Line 1\nLine 2")
```

```
Line 1
Line 2
```

```
1 print("Patrick\rLaub")
```

```
Laubick
```

```
1 print("C:\tom\new folder")
```

```
C:	om
ew folder
```

Escape the backslash:

```
1 print("C:\\tom\\new folder")
```

```
C:\tom\new folder
```

```
1 repr("Hello,\rworld!")
```

```
"'Hello,\\rworld!'"
```

Non-natural language processing I

How would you evaluate

> 10 + 2 * -3

All that Python sees is a string of characters.

```
1 [ord(c) for c in "10 + 2 * -3"]
```

```
[49, 48, 32, 43, 32, 50, 32, 42, 32, 45, 51]
```

```
1 10 + 2 * -3
```

```
4
```

Non-natural language processing II

Python first tokenizes the string:

```
1  code = "10 + 2 * -3"
2  tokens = tokenize.tokenize(io.BytesIO(code.encode("utf-8")).readline)
3  for token in tokens:
4      print(token)
```

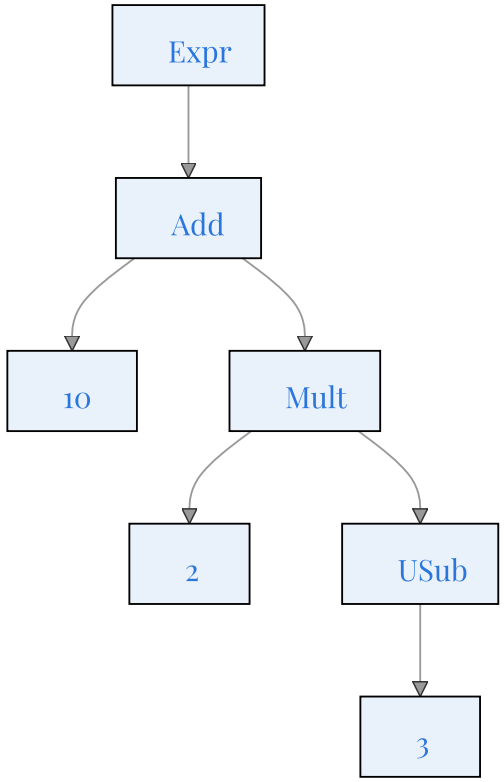
```
TokenInfo(type=68 (ENCODING), string='utf-8', start=(0, 0), end=(0, 0), line='')
TokenInfo(type=2 (NUMBER), string='10', start=(1, 0), end=(1, 2), line='10 + 2 * -3')
TokenInfo(type=55 (OP), string='+', start=(1, 3), end=(1, 4), line='10 + 2 * -3')
TokenInfo(type=2 (NUMBER), string='2', start=(1, 5), end=(1, 6), line='10 + 2 * -3')
TokenInfo(type=55 (OP), string='*', start=(1, 7), end=(1, 8), line='10 + 2 * -3')
TokenInfo(type=55 (OP), string='-', start=(1, 9), end=(1, 10), line='10 + 2 * -3')
TokenInfo(type=2 (NUMBER), string='3', start=(1, 10), end=(1, 11), line='10 + 2 * -3')
TokenInfo(type=4 (NEWLINE), string='', start=(1, 11), end=(1, 12), line='10 + 2 * -3')
TokenInfo(type=0 (ENDMARKER), string='', start=(2, 0), end=(2, 0), line='')
```

Non-natural language processing III

Python needs to *parse* the tokens into an abstract syntax tree.

```
1 print(ast.dump(ast.parse("10 + 2 * -3"), indent="  "))
```

```
Module(
  body=[
    Expr(
      value=BinOp(
        left=Constant(value=10),
        op=Add(),
        right=BinOp(
          left=Constant(value=2),
          op=Mult(),
          right=UnaryOp(
            op=USub(),
            operand=Constant(value=3)))))])
```



Non-natural language processing IV

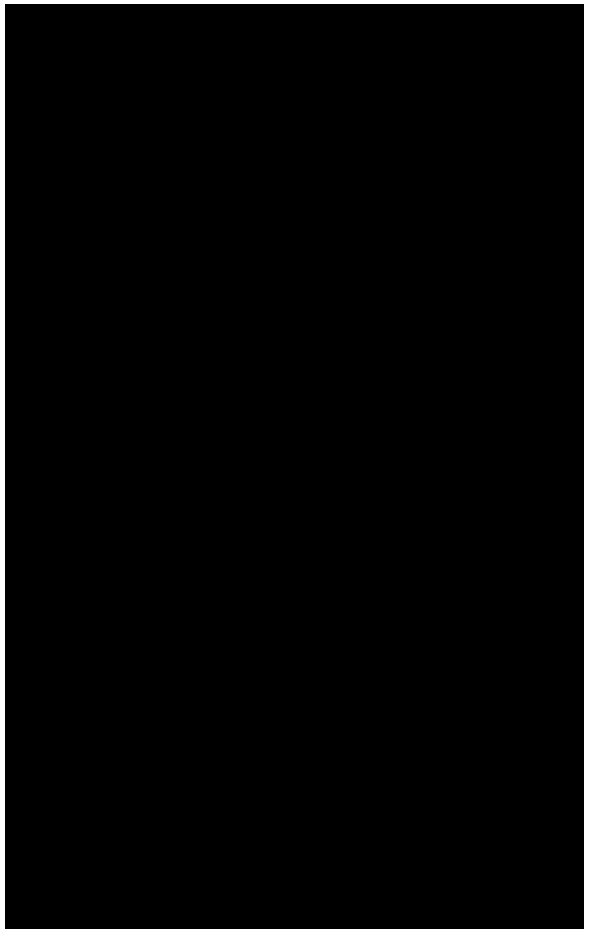
The abstract syntax tree is then compiled into bytecode.

```python
def expression(a, b, c):
    return a + b * -c

dis.dis(expression)
```

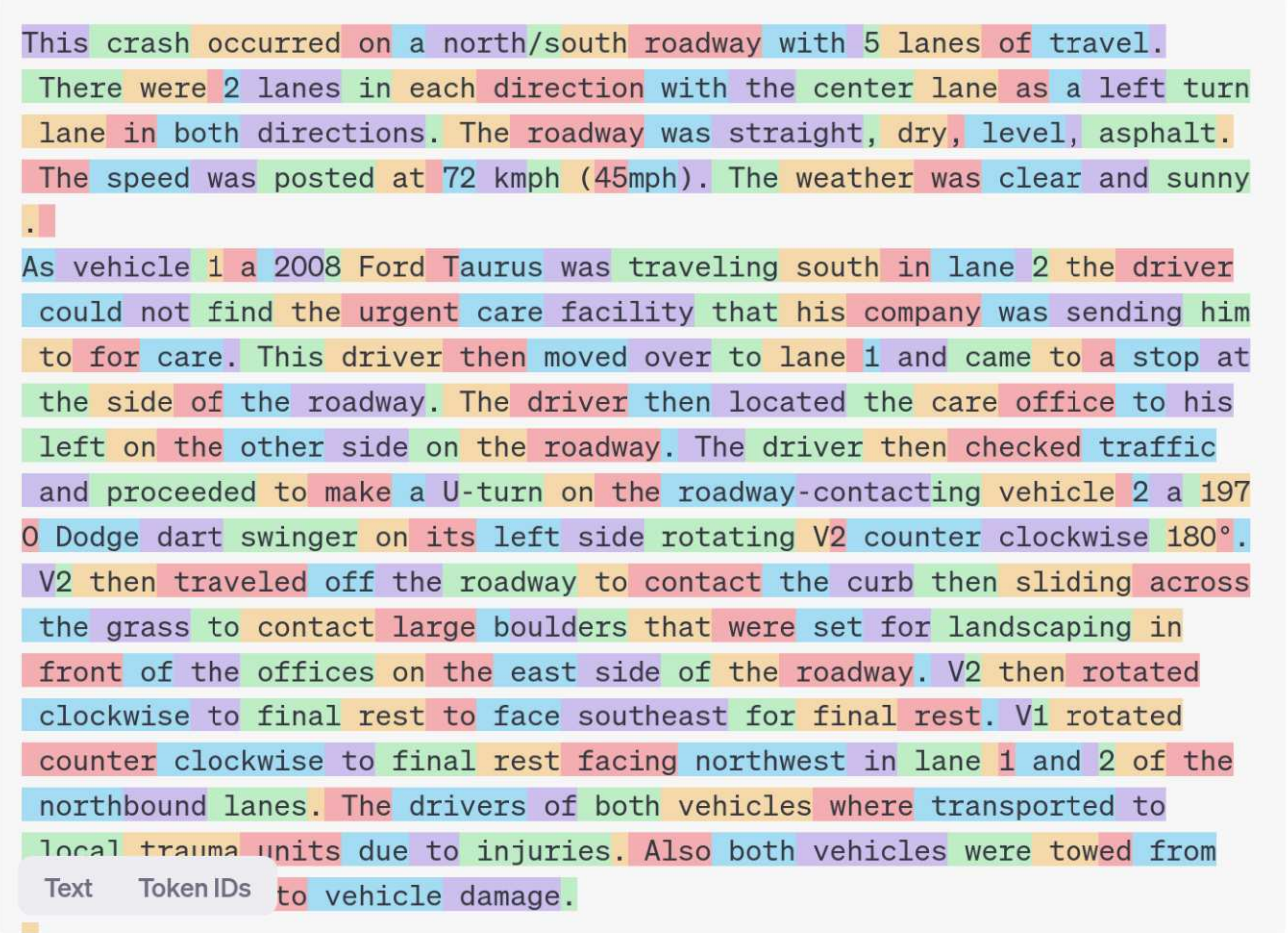
```
1           RESUME                   0

2           LOAD_FAST_BORROW_LOAD_FAST_BORROW 1 (a, b)
            LOAD_FAST_BORROW         2 (c)
            UNARY_NEGATIVE
            BINARY_OP                5 (*)
            BINARY_OP                0 (+)
            RETURN_VALUE
```



Running the bytecode

ChatGPT tokenization



Example of GPT 3.5/4's tokenization

E.g. radical for animals

- gǒu (dog)
- māo (cat)
- láng (wolf)
- shī (lion)

Source: https://platform.openai.com/tokenizer

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Downloading the dataset

Look at the (U.S.) National Highway Traffic Safety Administration's (NHTSA) National Motor Vehicle Crash Causation Survey (NMVCCS) dataset.

```
if not Path("data/NHTSA_NMVCCS_extract.parquet.gzip").exists():
    print("Downloading dataset")
    !wget -P data https://github.com/JSchelldorfer/ActuarialDataScience/raw/master/12%20-

df = pd.read_parquet("data/NHTSA_NMVCCS_extract.parquet.gzip")
print(f"shape of DataFrame: {df.shape}")
```

```
shape of DataFrame: (6949, 16)
```

Features

- `level_0`, `index`, `SCASEID`: all useless row numbers
- `SUMMARY_EN` and `SUMMARY_GE`: summaries of the accident
- `NUMTOTV`: total number of vehicles involved in the accident
- `WEATHER1` to `WEATHER8` (**not one-hot**):
  - `WEATHER1`: cloudy
  - `WEATHER2`: snow
  - `WEATHER3`: fog, smog, smoke
  - `WEATHER4`: rain
  - `WEATHER5`: sleet, hail (freezing drizzle or rain)
  - `WEATHER6`: blowing snow
  - `WEATHER7`: severe crosswinds
  - `WEATHER8`: other
- `INJSEVA` and `INJSEVB`: injury severity & (binary) presence of bodily injury

Source: JSchelldorfer's GitHub.

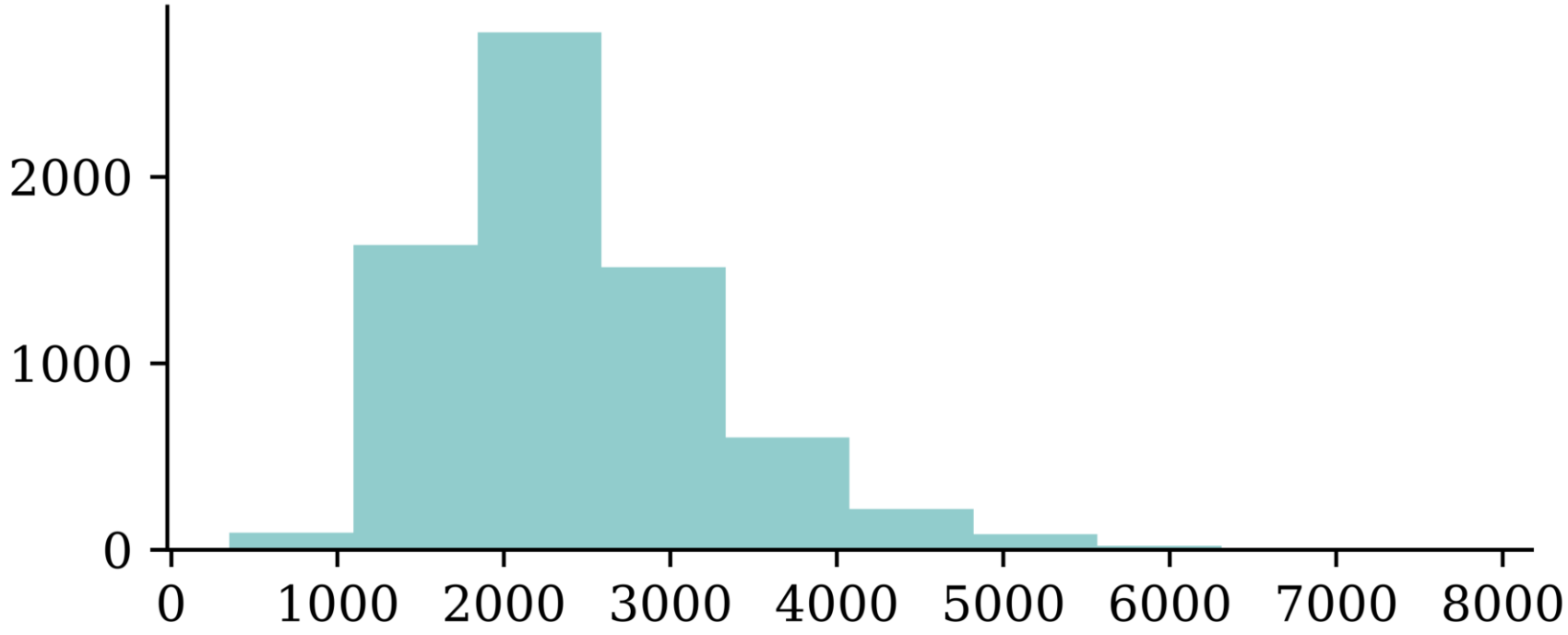
Crash summaries

```
1 df["SUMMARY_EN"]
```

```
0       V1, a 2000 Pontiac Montana minivan, made a lef ...
1       The crash occurred in the eastbound lane of a  ...
2       This crash occurred just after the noon time h ...
                                ...
6946    The crash occurred in the eastbound lanes of a ...
6947    This single-vehicle crash occurred in a rural  ...
6948    This two vehicle daytime collision occurred mi ...
Name: SUMMARY_EN, Length: 6949, dtype: str
```

The length of the summaries

```
1 df["SUMMARY_EN"].map(lambda summary: len(summary)).hist(grid=False);
```



A crash summary

```
1 df["SUMMARY_EN"].iloc[1]
```

"The crash occurred in the eastbound lane of a two-lane, two-way asphalt roadway on level grade. The conditions were daylight and wet with cloudy skies in the early afternoon on a weekday.\t\r \r V1, a 1995 Chevrolet Lumina was traveling eastbound. V2, a 2004 Chevrolet Trailblazer was also traveling eastbound on the same roadway. V2, was attempting to make a left-hand turn into a private drive on the North side of the roadway. While turning V1 attempted to pass V2 on the left-hand side contacting it's front to the left side of V2. Both vehicles came to final rest on the roadway at impact.\r \r The driver of V1 fled the scene and was not identified, so no further information could be obtained from him. The Driver of V2 stated that the driver was a male and had hit his head and was bleeding. She did not pursue the driver because she thought she saw a gun. The officer said that the car had been reported stolen.\r \r The Critical Precrash Event for the driver of V1 was this vehicle traveling over left lane line on the left side of travel. The Critical Reason for the Critical Event was coded as unknown reason for the critical event because the driver was not available. \r \r The driver of V2 was a 41-year old female who had reported that she had stopped prior to turning to make sure she was at the right house. She was going to show a house for a client. She had no health related problems. She had taken amoxicillin. She does not wear corrective lenses

Carriage returns

```
1 print(df["SUMMARY_EN"].iloc[1])
```

The Critical Precrash Event for the driver of V2 was other vehicle encroachment from adjacent lane over left lane line. The Critical Reason for the Critical Event was not coded for this vehicle and the driver of V2 was not thought to have contributed to the crash.r corrective lenses and felt rested. She was not injured in the crash. of V2. Both vehicles came to final rest on the roadway at impact.

```
1 # Replace every \r with \n
2 def replace_carriage_return(summary):
3     return summary.replace("\r", "\n")
4 
5 df["SUMMARY_EN"] = df["SUMMARY_EN"].map(replace_carriage_return)
6 print(df["SUMMARY_EN"].iloc[1][:500])
```

The crash occurred in the eastbound lane of a two-lane, two-way asphalt roadway on level grade. The conditions were daylight and wet with cloudy skies in the early afternoon on a weekday.

 V1, a 1995 Chevrolet Lumina was traveling eastbound. V2, a 2004 Chevrolet Trailblazer was also traveling eastbound on the same roadway. V2, was attempting to make a left-hand turn into a private drive on the North side of the roadway. While turning V1 attempted to pass V2 on the left-hand side contactin

Target

Predict number of vehicles in the crash.

```
df["NUMTOTV"].value_counts()\
    .sort_index()
```

```
NUMTOTV
1    1822
2    4151
3     783
4     150
5      34
6       5
7       2
8       1
9       1
Name: count, dtype: int64
```

```
np.sum(df["NUMTOTV"] > 3)
```

```
np.int64(193)
```

Simplify the target to just:

- 1 vehicle
- 2 vehicles
- 3+ vehicles

```
df["NUM_VEHICLES"] = \
  df["NUMTOTV"].map(lambda x: \
    str(x) if x ⩽ 2 else "3+")
df["NUM_VEHICLES"].value_counts()\
  .sort_index()
```

```
NUM_VEHICLES
1     1822
2     4151
3+     976
Name: count, dtype: int64
```

Give fake names to the vehicles

Find the words “V1”, “V2” and “V3” and replace them with random labels.

```
rnd.seed(123)

for i, summary in enumerate(df["SUMMARY_EN"]):
    word_numbers = ["one", "two", "three", "four", "five", "six", "seven", "eight", "nine
    num_cars = 10
    new_car_nums = [f"V{rnd.randint(100, 10000)}" for _ in range(num_cars)]
    num_spaces = 4

    for car in range(1, num_cars+1):
        new_num = new_car_nums[car-1]
        summary = summary.replace(f"V-{car}", new_num)
        summary = summary.replace(f"Vehicle {word_numbers[car-1]}", new_num).replace(f"ve
        summary = summary.replace(f"Vehicle #{word_numbers[car-1]}", new_num).replace(f"v
        summary = summary.replace(f"Vehicle {car}", new_num).replace(f"vehicle {car}", ne
        summary = summary.replace(f"Vehicle #{car}", new_num).replace(f"vehicle #{car}",
        summary = summary.replace(f"Vehicle # {car}", new_num).replace(f"vehicle # {car}"

        for j in range(num_spaces+1):
            summary = summary.replace(f"V{' '*j}{car}", new_num).replace(f"V{' '*j}#{car}
            summary = summary.replace(f"v{' '*j}{car}", new_num).replace(f"v{' '*j}#{car}

    df.loc[i, "SUMMARY_EN"] = summary
```

Convert y to integers & split the data

```
target_labels = df["NUM_VEHICLES"]
target = LabelEncoder().fit_transform(target_labels)
target
```

```
array([1, 1, 1, ..., 2, 0, 1], shape=(6949,))
```

```
weather_cols = [f"WEATHER{i}" for i in range(1, 9)]
features = df[["SUMMARY_EN"] + weather_cols]

X_main, X_test, y_main, y_test = \
    train_test_split(features, target, test_size=0.2, random_state=1)

# As 0.25 x 0.8 = 0.2
X_train, X_val, y_train, y_val = \
    train_test_split(X_main, y_main, test_size=0.25, random_state=1)

X_train.shape, X_val.shape, X_test.shape
```

```
((4169, 9), (1390, 9), (1390, 9))
```

```
print([np.mean(y_train == y) for y in [0, 1, 2]])
```

```
[np.float64(0.25833533221396016), np.float64(0.6032621731830176),
np.float64(0.1384024946030223)]
```

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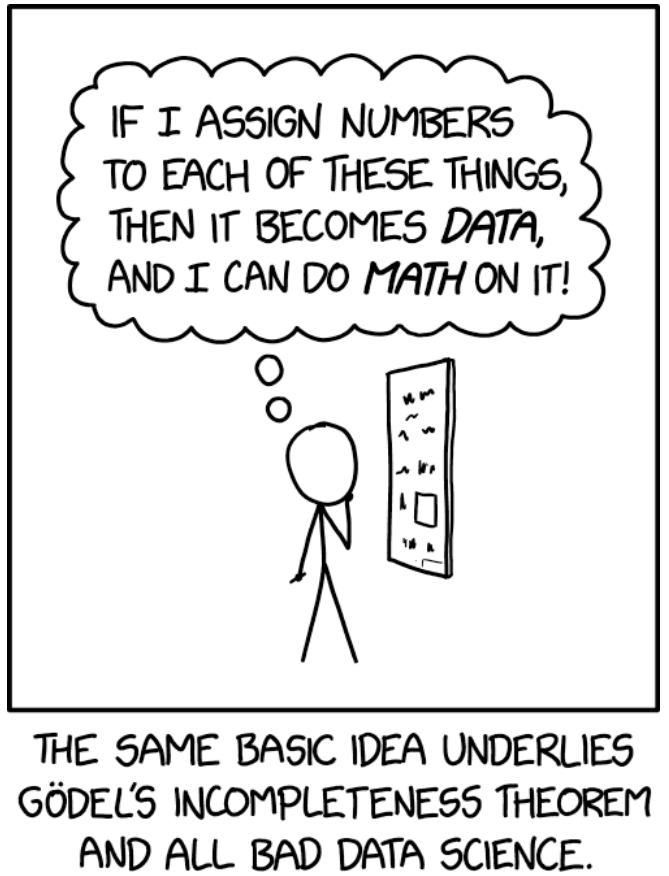
Text vectorisation

Text vectorisation describes the process of converting text into a numerical representation.

Popular methods for vectorisation include:

- Bag of words
- TF-IDF
- Word vectors



Assigning Numbers

Source: Randall Munroe (2022), xkcd #2610: Assigning Numbers.

Grab the start of a few summaries

```
1 first_summaries = X_train["SUMMARY_EN"].iloc[:3]
2 first_summaries
```

```
2532    This crash occurred in the early afternoon of  ...
6209    This two-vehicle crash occurred in a four-legg ...
2561    The crash occurred in the eastbound direction  ...
Name: SUMMARY_EN, dtype: str
```

```
1 first_words = first_summaries.map(lambda txt: txt.split(" ")[:7])
2 first_words
```

```
2532    [This, crash, occurred, in, the, early, aftern ...
6209    [This, two-vehicle, crash, occurred, in, a, fo ...
2561    [The, crash, occurred, in, the, eastbound, dir ...
Name: SUMMARY_EN, dtype: object
```

```
1 start_of_summaries = first_words.map(lambda txt: " ".join(txt))
2 start_of_summaries
```

```
2532          This crash occurred in the early afternoon
6209    This two-vehicle crash occurred in a four-legged
2561       The crash occurred in the eastbound direction
Name: SUMMARY_EN, dtype: str
```

Count words in the first summaries

```
1 vect = CountVectorizer()
2 counts = vect.fit_transform(start_of_summaries)
3 vocab = vect.get_feature_names_out()
4 print(len(vocab), vocab)
```

```
13 ['afternoon' 'crash' 'direction' 'early' 'eastbound' 'four' 'in' 'legged'
 'occurred' 'the' 'this' 'two' 'vehicle']
```

```
1 counts
```

```
<Compressed Sparse Row sparse matrix of dtype 'int64'
    with 21 stored elements and shape (3, 13)>
```

```
1 counts.toarray()
```

```
array([[1, 1, 0, 1, 0, 0, 1, 0, 1, 1, 1, 0, 0],
       [0, 1, 0, 0, 0, 1, 1, 1, 1, 0, 1, 1, 1],
       [0, 1, 1, 0, 1, 0, 1, 0, 1, 2, 0, 0, 0]])
```

Encode new sentences to BoW

```
vect.transform([
    "first car hit second car in a crash",
    "ios 27 beta released",
])
```

```
<Compressed Sparse Row sparse matrix of dtype 'int64'
    with 2 stored elements and shape (2, 13)>
```

```
vect.transform([
    "first car hit second car in a crash",
    "ios 27 beta released",
]).toarray()
```

```
array([[0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]])
```

```
print(vocab)
```

```
['afternoon' 'crash' 'direction' 'early' 'eastbound' 'four' 'in' 'legged'
 'occurred' 'the' 'this' 'two' 'vehicle']
```

Bag of n-grams

```
1  vect = CountVectorizer(ngram_range=(1, 2))
2  counts = vect.fit_transform(start_of_summaries)
3  vocab = vect.get_feature_names_out()
4  print(len(vocab), vocab)
```

```
27 ['afternoon' 'crash' 'crash occurred' 'direction' 'early'
 'early afternoon' 'eastbound' 'eastbound direction' 'four' 'four legged'
 'in' 'in four' 'in the' 'legged' 'occurred' 'occurred in' 'the'
 'the crash' 'the early' 'the eastbound' 'this' 'this crash' 'this two'
 'two' 'two vehicle' 'vehicle' 'vehicle crash']
```

```
1  counts.toarray()
```

```
array([[1, 1, 1, 0, 1, 1, 0, 0, 0, 0, 1, 0, 1, 0, 1, 1, 1, 0, 1, 0, 1, 1,
        0, 0, 0, 0, 0],
       [0, 1, 1, 0, 0, 0, 0, 0, 1, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 0, 1, 0,
        1, 1, 1, 1, 1],
       [0, 1, 1, 1, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1, 1, 2, 1, 0, 1, 0, 0,
        0, 0, 0, 0, 0]])
```

See: Google Books Ngram Viewer

Count words in all the summaries

```
vect = CountVectorizer()
vect.fit(X_train["SUMMARY_EN"])
vocab = list(vect.get_feature_names_out())
len(vocab)
```

```
18866
```

```
vocab[:5], vocab[len(vocab)//2:(len(vocab)//2 + 5)], vocab[-5:]
```

```
(['00', '000', '000lbs', '003', '005'],
 ['swinger', 'swinging', 'swipe', 'swiped', 'swiping'],
 ['zorcor', 'zotril', 'zx2', 'zx5', 'zyrtec'])
```

Create the X matrices

```python
def vectorise_dataset(X, vect, txt_col="SUMMARY_EN", dataframe=False):
    X_vects = vect.transform(X[txt_col]).toarray()
    X_other = X[[f"WEATHER{i}" for i in range(1, 9)]]

    if not dataframe:
        return np.concatenate([X_vects, X_other], axis=1)
    else:
        # Add column names and indices to the combined dataframe.
        vocab = list(vect.get_feature_names_out())
        X_vects_df = pd.DataFrame(X_vects, columns=vocab, index=X.index)
        return pd.concat([X_vects_df, X_other], axis=1)
```

```python
X_train_bow = vectorise_dataset(X_train, vect)
X_val_bow = vectorise_dataset(X_val, vect)
X_test_bow = vectorise_dataset(X_test, vect)
```

Check the input matrix

```
1 vectorise_dataset(X_train, vect, dataframe=True)
```

	00	000	000lbs	003	005	007	00am	00pm	00tyd02	01	...	zx5
2532	0	0	0	0	0	0	0	0	0	0	...	0
6209	0	0	0	0	0	0	0	0	0	0	...	0
2561	0	0	0	0	0	0	0	0	0	0	...	0
...	...	...	...	...	...	...	...	...	...	...	...	...
6882	0	0	0	0	0	0	0	0	0	0	...	0
206	0	0	0	0	0	0	0	0	0	0	...	0
6356	0	0	0	0	0	0	0	0	0	0	...	0

4169 rows × 18874 columns

Make a simple dense model

```
num_features = X_train_bow.shape[1]
num_cats = 3 # 1, 2, 3+ vehicles

def build_model(num_features, num_cats):
    random.seed(42)

    model = Sequential([
        Input((num_features,)),
        Dense(100, activation="relu"),
        Dense(num_cats, activation="softmax")
    ])

    topk = SparseTopKCategoricalAccuracy(k=2, name="topk")
    model.compile("adam", "sparse_categorical_crossentropy",
        metrics=["accuracy", topk])

    return model
```

Inspect the model

```
model = build_model(num_features, num_cats)
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 100)	1,887,500
dense_1 (Dense)	(None, 3)	303

Total params: 1,887,803 (7.20 MB)
Trainable params: 1,887,803 (7.20 MB)
Non-trainable params: 0 (0.00 B)

Fit & evaluate the model

```
es = EarlyStopping(patience=1, restore_best_weights=True,
    monitor="val_accuracy", verbose=2)
%time hist = model.fit(X_train_bow, y_train, epochs=10, \
    callbacks=[es], validation_data=(X_val_bow, y_val), verbose=0);
```

```
Epoch 2: early stopping
Restoring model weights from the end of the best epoch: 1.
CPU times: user 1.36 s, sys: 197 ms, total: 1.56 s
Wall time: 725 ms
```

```
model.evaluate(X_train_bow, y_train, verbose=0)
```

```
[0.05817717686295509, 0.9901655316352844, 0.9995202422142029]
```

```
model.evaluate(X_val_bow, y_val, verbose=0)
```

```
[0.1902538239955902, 0.9503597021102905, 0.9942445755004883]
```

Lecture Outline

- Natural Language Processing
- How Computers View Text
- Car Crash Police Reports
- Text Vectorisation with Bag of Words
- **Discard Infrequent Words and Stop Words**
- Merge Together Similar Words
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- Demo of Word Embeddings
- Recap

The `max_features` value

```
vect = CountVectorizer(max_features=10)
vect.fit(X_train["SUMMARY_EN"])
vocab = vect.get_feature_names_out()
len(vocab)
```

```
10
```

```
print(vocab)
```

```
['and' 'driver' 'for' 'in' 'lane' 'of' 'the' 'to' 'vehicle' 'was']
```

What is left?

```python
for i in range(3):
    sentence = X_train["SUMMARY_EN"].iloc[i]
    for word in sentence.split(" ")[:10]:
        word_or_qn = word if word in vocab else "?"
        print(word_or_qn, end=" ")
    print() # Same as print("\n", end="")
```

```
? ? ? in the ? ? of ? ?
? ? ? ? in ? ? ? ? ?
? ? ? in the ? ? of ? ?
```

```python
for i in range(3):
    sentence = X_train["SUMMARY_EN"].iloc[i]
    num_words = 0
    for word in sentence.split(" "):
        if word in vocab:
            print(word, end=" ")
            num_words += 1
        if num_words == 10:
            break
    print()
```

```
in the of in the of of was and was
in and of in and for the of the and
in the of to was was of was was and
```

Remove stop words

```
vect = CountVectorizer(max_features=10, stop_words="english")
vect.fit(X_train["SUMMARY_EN"])
vocab = vect.get_feature_names_out()
len(vocab)
```

```
10
```

```
print(vocab)
```

```
['coded' 'crash' 'critical' 'driver' 'event' 'intersection' 'lane' 'left'
 'roadway' 'vehicle']
```

```
for i in range(3):
    sentence = X_train["SUMMARY_EN"].iloc[i]
    num_words = 0
    for word in sentence.split(" "):
        if word in vocab:
            print(word, end=" ")
            num_words += 1
        if num_words == 10:
            break
    print()
```

```
crash intersection roadway roadway roadway intersection lane lane intersection driver
crash roadway left roadway roadway roadway lane lane roadway crash
crash vehicle left left vehicle driver vehicle lane lane coded
```

Keep 1,000 most frequent words

```
vect = CountVectorizer(max_features=1_000, stop_words="english")
vect.fit(X_train["SUMMARY_EN"])
vocab = vect.get_feature_names_out()
len(vocab)
```

```
1000
```

```
print(vocab[:5], vocab[len(vocab)//2:(len(vocab)//2 + 5)], vocab[-5:])
```

```
['10' '105' '113' '12' '15'] ['interruption' 'intersected' 'intersecting' 'intersection'
'interstate'] ['year' 'years' 'yellow' 'yield' 'zone']
```

Create the X matrices:

```
X_train_bow = vectorise_dataset(X_train, vect)
X_val_bow = vectorise_dataset(X_val, vect)
X_test_bow = vectorise_dataset(X_test, vect)
```

What is left?

```python
for i in range(8):
    sentence = X_train["SUMMARY_EN"].iloc[i]
    num_words = 0
    for word in sentence.split(" "):
        if word in vocab:
            print(word, end=" ")
            num_words += 1
        if num_words == 10:
            break
    print()
```

```
crash occurred early afternoon weekday middle suburban intersection consisted lanes
crash occurred roadway level consists lanes direction center left turn
crash occurred eastbound direction entrance ramp right curved road uphill
crash occurred straight roadway consists lanes direction center left turn
collision occurred evening hours crash occurred level bituminous roadway residential
vehicle crash occurred daylight location lane undivided left curved downhill
vehicle crash occurred early morning daylight hours roadway traffic roadway
crash occurred northbound lanes northbound southbound slightly street curved posted
```

Note

We hope to see SMS-like language, with limited vocabulary but still able to understand it.

Check the input matrix

```
1 vectorise_dataset(X_train, vect, dataframe=True)
```

	10	105	113	12	15	150	16	17	18	180	...	yield	zone	WEATHER1	V
2532	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0
6209	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0
2561	1	0	1	0	0	0	0	0	0	0	...	0	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
6882	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0
206	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0
6356	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0

4169 rows × 1008 columns

Make & inspect the model

```
num_features = X_train_bow.shape[1]
model = build_model(num_features, num_cats)
model.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
dense_2 (Dense)	(None, 100)	100,900
dense_3 (Dense)	(None, 3)	303

Total params: 101,203 (395.32 KB)
Trainable params: 101,203 (395.32 KB)
Non-trainable params: 0 (0.00 B)

Fit & evaluate the model

```
es = EarlyStopping(patience=1, restore_best_weights=True,
    monitor="val_accuracy", verbose=2)
%time hist = model.fit(X_train_bow, y_train, epochs=10, \
    callbacks=[es], validation_data=(X_val_bow, y_val), verbose=0);
```

```
Epoch 2: early stopping
Restoring model weights from the end of the best epoch: 1.
CPU times: user 271 ms, sys: 121 ms, total: 392 ms
Wall time: 293 ms
```

```
model.evaluate(X_train_bow, y_train, verbose=0)
```

```
[0.1888578087091446, 0.9549052715301514, 0.9980810880661011]
```

```
model.evaluate(X_val_bow, y_val, verbose=0)
```

```
[0.265645831823349, 0.9258992671966553, 0.9942445755004883]
```

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- Recap

Keep 1,000 most frequent words

```
vect = CountVectorizer(max_features=1_000, stop_words="english")
vect.fit(X_train["SUMMARY_EN"])
vocab = vect.get_feature_names_out()
len(vocab)
```

```
1000
```

```
print(vocab[:5], vocab[len(vocab)//2:(len(vocab)//2 + 5)], vocab[-5:])
```

```
['10' '105' '113' '12' '15'] ['interruption' 'intersected' 'intersecting' 'intersection'
'interstate'] ['year' 'years' 'yellow' 'yield' 'zone']
```

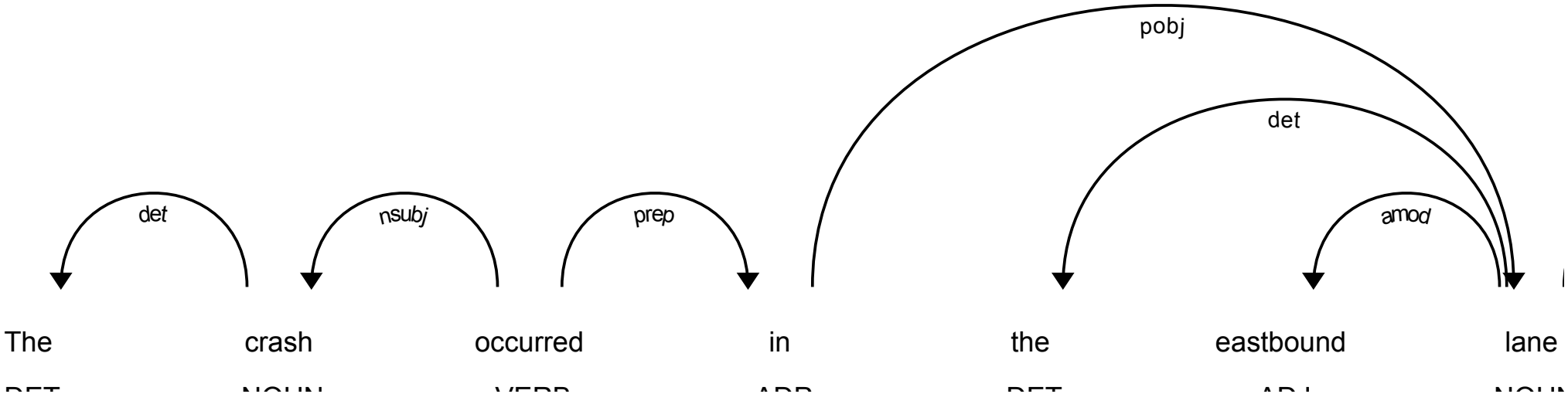
Use some knowledge of the English language

```
1 nlp = spacy.load("en_core_web_trf")
2 doc = nlp("Apple is looking at buying U.K. startup for $1 billion")
3 for token in doc:
4     print(token.text, token.pos_, token.dep_, token.lemma_)
```

```
Apple PROPN nsubj Apple
is AUX aux be
looking VERB ROOT look
at ADP prep at
buying VERB pcomp buy
U.K. PROPN compound U.K.
startup NOUN dobj startup
for ADP prep for
$ SYM quantmod $
1 NUM compound 1
billion NUM pobj billion
```

Dependency visualiser

```
doc = nlp(df["SUMMARY_EN"].iloc[1])
spacy.displacy.render(doc, style="dep")
```



Entity recognition

```
1 doc = nlp(df["SUMMARY_EN"].iloc[1])
2 spacy.displacy.render(doc, style="ent")
```

The crash occurred in the eastbound lane of a two CARDINAL -lane, two CARDINAL -way asphalt roadway on level grade. The conditions were daylight and wet with cloudy skies in the early afternoon TIME on a weekday DATE .

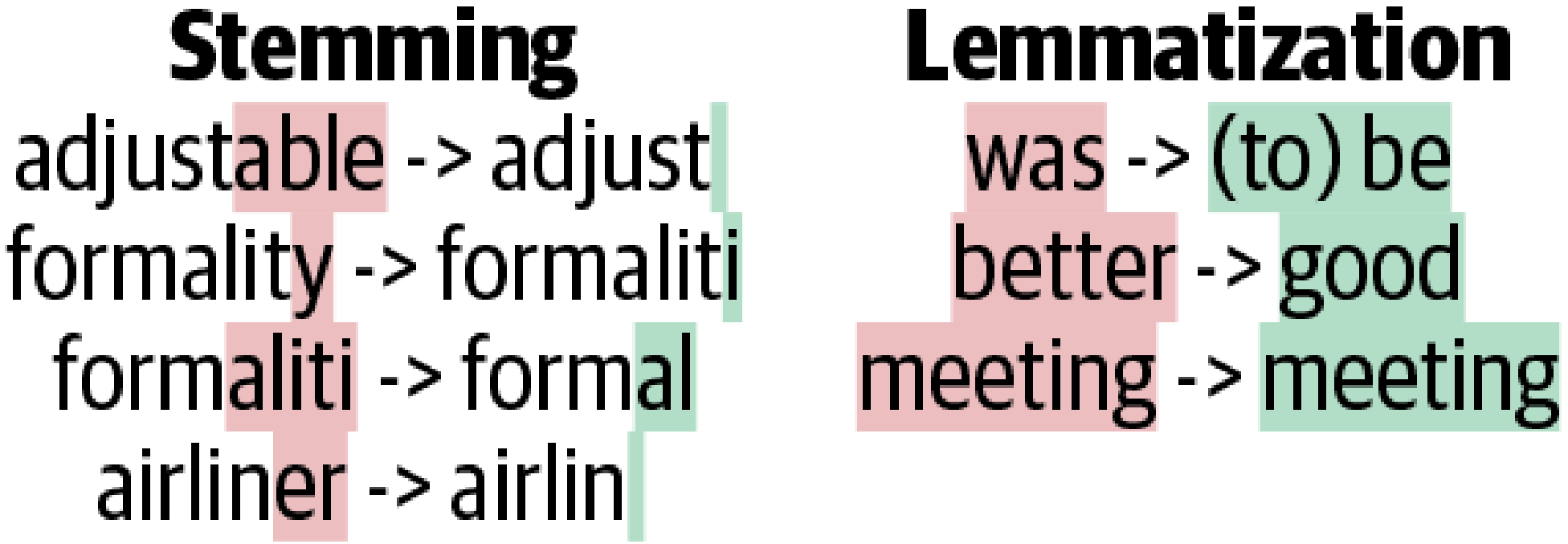
V342542243 PRODUCT , a 1995 DATE Chevrolet ORG Lumina PRODUCT was traveling eastbound. V342542269 PRODUCT , a 2004 DATE Chevrolet ORG Trailblazer PRODUCT was also traveling eastbound on the same roadway. V342542269 PRODUCT , was attempting to make a left-hand turn into a private drive on the North side of the roadway. While turning V342542243 PRODUCT attempted to pass V342542269 PRODUCT on the left-hand side contacting it's front to the left side of V342542269 PRODUCT . Both vehicles came to final rest on the roadway at impact.

The driver of V342542243 PRODUCT fled the scene and was not identified, so no further information could be obtained from him. The Driver of V342542269 PRODUCT stated that the driver was a male and had hit his head and was bleeding. She did not pursue the driver because she thought she saw a gun. The officer said that the car had been reported stolen.

Stemming and lemmatizing

> "Stemming refers to the process of removing suffixes and reducing a word to some base form such that all different variants of that word can be represented by the same form (e.g., "car" and "cars" are both reduced to "car"). This is accomplished by applying a fixed set of rules (e.g., if the word ends in "-es," remove "-es"). More such examples are shown in Figure 2-7. Although such rules may not always end up in a linguistically correct base form, stemming is commonly used in search engines to match user queries to relevant documents and in text classification to reduce the feature space to train machine learning models.
>
> Lemmatization is the process of mapping all the different forms of a word to its base word, or lemma. While this seems close to the definition of stemming, they are, in fact, different. For example, the adjective "better," when stemmed, remains the same. However, upon lemmatization, this should become "good," as shown in Figure 2-7. Lemmatization requires more linguistic knowledge, and modeling and developing efficient lemmatizers remains an open problem in NLP research even now."
>
> — Vajjala et al. (2020)

Stemming and lemmatizing examples



Examples of stemming and lemmatization

Original: “The striped bats are hanging on their feet for best”

Stemmed: “the stripe bat are hang on their feet for best”

Lemmatized: “the stripe bat be hang on their foot for good”

Source: Kushwah (2019) What is difference between stemming and lemmatization?, Quora.

Examples

Stemmed

organization ↠ organ

civilization ↠ civil

information ↠ inform

consultant ↠ consult

Lemmatized

Here's looking at you, kid. ↠ here be look at you , kid .

Lemmatize the text

```
def lemmatize(txt):
    doc = nlp(txt)
    good_tokens = [token.lemma_.lower() for token in doc \
        if not token.like_num and \
           not token.is_punct and \
           not token.is_space and \
           not token.is_currency and \
           not token.is_stop]
    return " ".join(good_tokens)
```

```
test_str = "Incident at 100kph and '10 incidents -13.3%' are incidental?\t $5"
lemmatize(test_str)
```

```
'incident 100kph incident incidental'
```

```
test_str = "I interviewed 5-years ago, 150 interviews every year at 10:30 are.."
lemmatize(test_str)
```

```
'interview year ago interview year 10:30'
```

Apply to the whole dataset

```
df["SUMMARY_EN_LEMMA"] = df["SUMMARY_EN"].map(lemmatize)
```

```
# The lemma column was computed after the train/val/test split, so attach it to
# the existing sets (same rows, so y_train/y_val/y_test still apply).
X_train["SUMMARY_EN_LEMMA"] = df.loc[X_train.index, "SUMMARY_EN_LEMMA"]
X_val["SUMMARY_EN_LEMMA"] = df.loc[X_val.index, "SUMMARY_EN_LEMMA"]
X_test["SUMMARY_EN_LEMMA"] = df.loc[X_test.index, "SUMMARY_EN_LEMMA"]

X_train.shape, X_val.shape, X_test.shape
```

```
((4169, 10), (1390, 10), (1390, 10))
```

What is left?

```
1 print("Original:", df["SUMMARY_EN"].iloc[0][:250])
```

```
Original: V6357885318682, a 2000 Pontiac Montana minivan, made a left turn from a private driveway onto a northbound 5-lane two-way, dry asphalt roadway on a downhill grade.  The posted speed limit on this roadway was 80 kmph (50 MPH). V6357885318682 entered t
```

```
1 print("Lemmatized:", df["SUMMARY_EN_LEMMA"].iloc[0][:250])
```

```
Lemmatized: v6357885318682 pontiac montana minivan left turn private driveway northbound lane way dry asphalt roadway downhill grade post speed limit roadway kmph mph v6357885318682 enter roadway cross southbound lane enter northbound lane left turn lane way int
```

```
1 print("Original:", df["SUMMARY_EN"].iloc[1][:250])
```

```
Original: The crash occurred in the eastbound lane of a two-lane, two-way asphalt roadway on level grade.  The conditions were daylight and wet with cloudy skies in the early afternoon on a weekday.

 V342542243, a 1995 Chevrolet Lumina was traveling eastbou
```

```
1 print("Lemmatized:", df["SUMMARY_EN_LEMMA"].iloc[1][:250])
```

```
Lemmatized: crash occur eastbound lane lane way asphalt roadway level grade condition daylight wet cloudy sky early afternoon weekday v342542243 chevrolet lumina travel eastbound v342542269 chevrolet trailblazer travel eastbound roadway v342542269 attempt left h
```

UNSW SYDNEY

Keep 1,000 most frequent lemmas

```
vect = CountVectorizer(max_features=1_000, stop_words="english")
vect.fit(X_train["SUMMARY_EN_LEMMA"])
vocab = vect.get_feature_names_out()
len(vocab)
```

```
1000
```

```
print(vocab[:5], vocab[len(vocab)//2:(len(vocab)//2 + 5)], vocab[-5:])
```

```
['10' '150' '48kmph' '4×4' '56kmph'] ['lens' 'lesabre' 'let' 'level' 'lexus'] ['yaw' 'year'
'yellow' 'yield' 'zone']
```

Create the X matrices:

```
X_train_bow = vectorise_dataset(X_train, vect, "SUMMARY_EN_LEMMA")
X_val_bow = vectorise_dataset(X_val, vect, "SUMMARY_EN_LEMMA")
X_test_bow = vectorise_dataset(X_test, vect, "SUMMARY_EN_LEMMA")
```

Check the input matrix

```
1 vectorise_dataset(X_train, vect, "SUMMARY_EN_LEMMA", dataframe=True)
```

	10	150	48kmph	4x4	56kmph	64kmph	72kmph	ability	able	acceler
2532	0	0	0	0	0	0	0	0	0	0
6209	0	0	0	0	0	0	0	0	0	0
2561	0	0	0	0	1	1	0	0	0	0
...	...	...	...	...	...	...	...	...	...	...
6882	0	0	0	0	0	0	0	0	0	0
206	0	0	0	0	0	0	0	0	0	0
6356	0	0	0	0	0	0	0	0	0	0

4169 rows × 1008 columns

Make & inspect the model

```
num_features = X_train_bow.shape[1]
model = build_model(num_features, num_cats)
model.summary()
```

Model: "sequential_2"

Layer (type)	Output Shape	Param #
dense_4 (Dense)	(None, 100)	100,900
dense_5 (Dense)	(None, 3)	303

Total params: 101,203 (395.32 KB)
Trainable params: 101,203 (395.32 KB)
Non-trainable params: 0 (0.00 B)

Fit & evaluate the model

```
es = EarlyStopping(patience=1, restore_best_weights=True,
    monitor="val_accuracy", verbose=2)
%time hist = model.fit(X_train_bow, y_train, epochs=10, \
    callbacks=[es], validation_data=(X_val_bow, y_val), verbose=0);
```

```
Epoch 4: early stopping
Restoring model weights from the end of the best epoch: 3.
CPU times: user 531 ms, sys: 236 ms, total: 767 ms
Wall time: 573 ms
```

```
model.evaluate(X_train_bow, y_train, verbose=0)
```

```
[0.06072762981057167, 0.9920844435691833, 0.9997601509094238]
```

```
model.evaluate(X_val_bow, y_val, verbose=0)
```

```
[0.18185271322727203, 0.9467625617980957, 0.9942445755004883]
```

Lecture Outline

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Term Frequency-Inverse Document Frequency

$$w_{x,y} = tf_{x,y} \times \log\left(\frac{N}{df_x}\right)$$

TF-IDF

Term x within document y

$tf_{x,y}$ = frequency of x in y

df_x = number of documents containing x

N = total number of documents

Infographic explaining TF-IDF

Source: FiloTechnologia (2014), A simple Java class for TF-IDF scoring, Blog post.

Using TF-IDF vectorisation

```
1 vect = TfidfVectorizer(max_features=1_000, stop_words="english")
2 vect.fit(X_train["SUMMARY_EN"])
3
4 X_train_tfidf = vectorise_dataset(X_train, vect)
5 X_val_tfidf = vectorise_dataset(X_val, vect)
6 X_test_tfidf = vectorise_dataset(X_test, vect)
7
8 vocab = vect.get_feature_names_out()
9 print(list(vocab[:50]))
```

```
['10', '105', '113', '12', '15', '150', '16', '17', '18', '180', '19', '1988', '1989', '1990',
'1991', '1992', '1993', '1994', '1995', '1996', '1997', '1998', '1999', '20', '2000', '2001',
'2002', '2003', '2004', '2005', '2006', '2007', '21', '22', '23', '24', '25', '26', '27',
'28', '29', '30', '30mph', '31', '32', '33', '34', '35', '35mph', '36']
```

The TF-IDF vectors

```
1 vectorise_dataset(X_train, vect, dataframe=True)
```

	10	105	113	12	15	150	16	17	18	180	...	yield	zo
2532	0.000000	0.0	0.00000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.(
6209	0.000000	0.0	0.00000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.(
2561	0.058082	0.0	0.08834	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.(
...	...	...	...	...	...	...	...	...	...	...	...	...	...
6882	0.000000	0.0	0.00000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.(
206	0.000000	0.0	0.00000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.(
6356	0.000000	0.0	0.00000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.(

4169 rows × 1008 columns

Feed TF-IDF into an ANN

```
num_features = X_train_tfidf.shape[1]
tfidf_model = build_model(num_features, num_cats)
tfidf_model.summary()
```

Model: "sequential_3"

Layer (type)	Output Shape	Param #
dense_6 (Dense)	(None, 100)	100,900
dense_7 (Dense)	(None, 3)	303

Total params: 101,203 (395.32 KB)
Trainable params: 101,203 (395.32 KB)
Non-trainable params: 0 (0.00 B)

Fit & evaluate

```
es = EarlyStopping(patience=10, restore_best_weights=True,
    monitor="val_accuracy", verbose=2)
tfidf_model.fit(X_train_tfidf, y_train, epochs=1_000, callbacks=[es],
    validation_data=(X_val_tfidf, y_val), verbose=0)
```

```
tfidf_model.evaluate(X_train_tfidf, y_train, verbose=0, batch_size=1_000)
```

```
[0.04896169528365135, 0.9966418743133545, 1.0]
```

```
tfidf_model.evaluate(X_val_tfidf, y_val, verbose=0, batch_size=1_000)
```

```
[0.1829308122396469, 0.9374100565910339, 0.9935252070426941]
```

Comparing the models

Same dense network throughout — only the **text representation** changes.

Model	Features	Parameters	Train accuracy	Val. accuracy	Val. loss	Val. top-2
Bag of words (full vocab)	18,874	1,887,803	99.0%	95.0%	0.190	99.4%
Lemmatized BoW (top 1,000)	1,008	101,203	99.2%	94.7%	0.182	99.4%
TF-IDF (top 1,000)	1,008	101,203	99.7%	93.7%	0.183	99.4%
Bag of words (top 1,000)	1,008	101,203	95.5%	92.6%	0.266	99.4%

Lecture Outline

- Natural Language Processing
- How Computers View Text
- Car Crash Police Reports
- Text Vectorisation with Bag of Words
- Discard Infrequent Words and Stop Words
- Merge Together Similar Words
- Use a Continuous Representation
- **Word Embeddings**
- Demo of Word Embeddings
- Recap

Each column used to be a word

Both bag-of-words and TF-IDF give a *column for every word* in the vocabulary, so you can always read off what a number refers to:

Representation	**Values**	**What the "crash" column holds**
Bag of words	integers $0, 1, 2, \ldots$	number of times "crash" appears
TF-IDF	continuous, ≥ 0	down-weighted count of "crash"

TF-IDF made the numbers *continuous*, but each column still *means one specific word*.

Word embeddings: continuous but unlabelled

Word embeddings (Word2Vec, GloVe) are continuous too, but the resemblance stops there:

- the vector is *short & dense* — e.g. 300 numbers, almost all non-zero;
- the columns are *learned dimensions*, not words.

So a word becomes something like

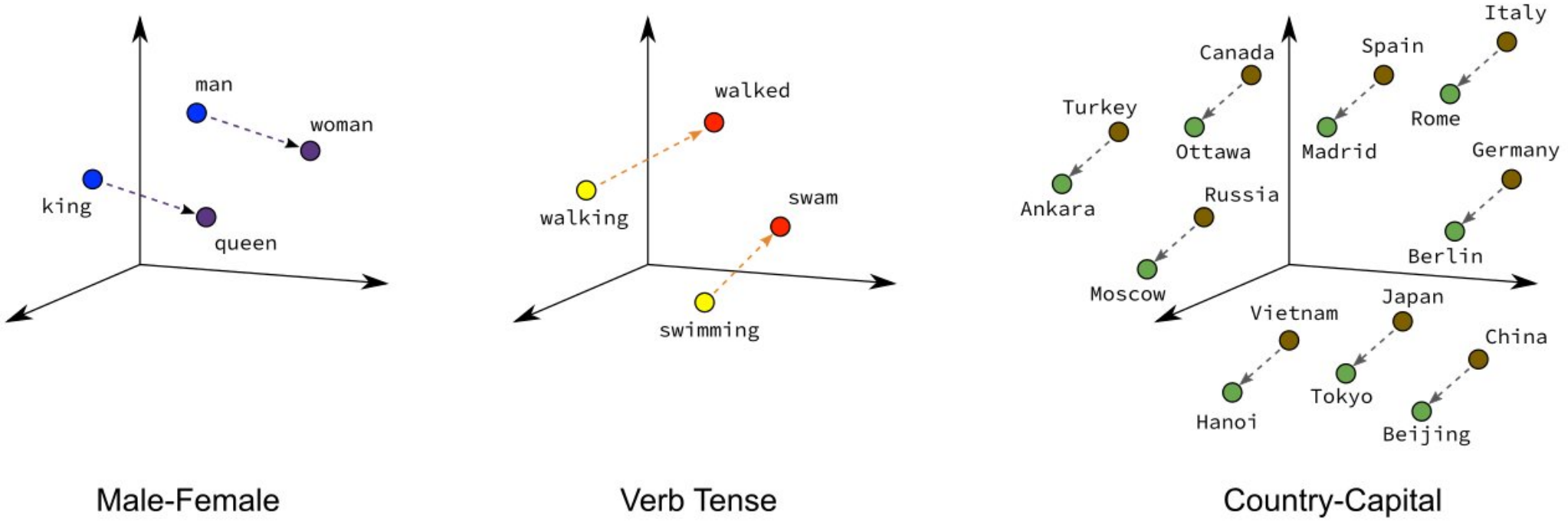
$$\text{crash} \longrightarrow [0.07,\ -0.42,\ 0.31,\ \ldots,\ -0.05]$$

and there is *no way to say what dimension 137 means*. The meaning is spread across the *whole* vector — it lives in the directions and distances between words, not in any single column.

Word Vectors

- One-hot representations capture word ‘existence’ only, whereas word vectors capture information about word meaning as well as location.
- This enables deep learning NLP models to automatically learn linguistic features.
- *Word2Vec & GloVe* are popular algorithms for generating word embeddings (i.e. word vectors).

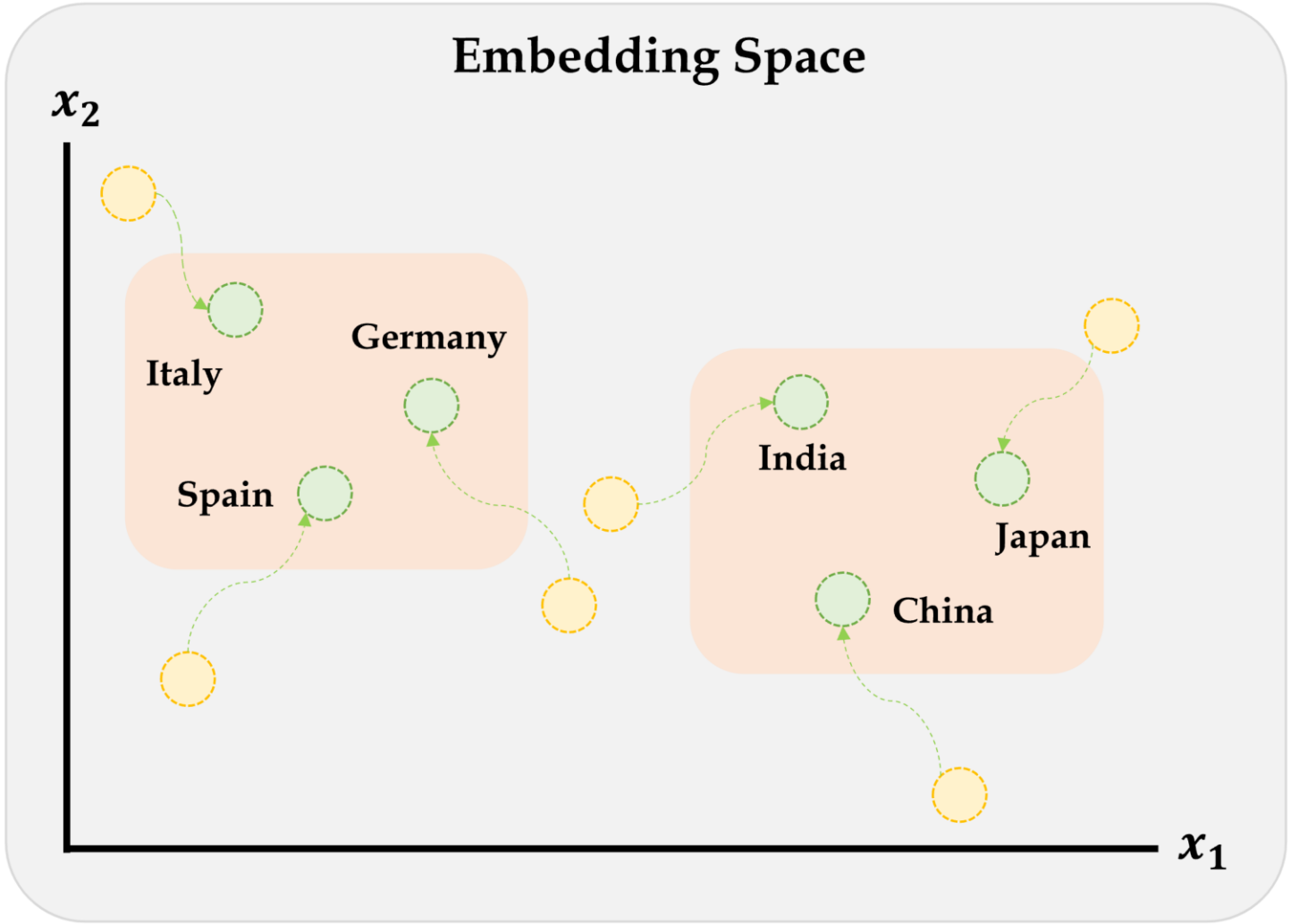
Word Vectors II



Illustrative word vectors.

Bogacz (2023), *Text Vectorization, an Introduction*, The Information Lab.

Illustration of the embeddings being learned



Embeddings will gradually improve during training.

Source: Marcus Lautier (2022).

Word2Vec

Key idea: You're known by the company you keep.

Two algorithms are used to calculate embeddings:

- *Continuous bag of words*: uses the context words to predict the target word
- *Skip-gram*: uses the target word to predict the context words

Predictions are made using a neural network with one hidden layer. Through backpropagation, we update a set of "weights" which become the word vectors.

Paper: Mikolov et al. (2013).

Word2Vec training methods

Continuous bag of words is a *centre word prediction* task

Skip-gram is a *neighbour word prediction* task

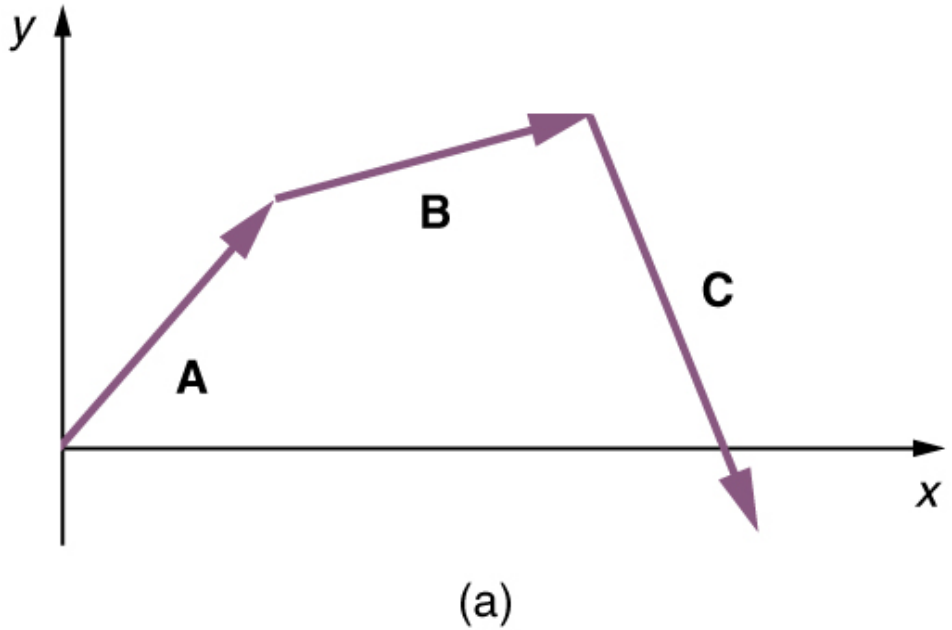
Suggested viewing

Computerphile (2019), Vectoring Words (Word Embeddings), YouTube (16 mins).

Source: Amit Chaudhary (2020), Self Supervised Representation Learning in NLP.

Word Vector Arithmetic

Relationships between words becomes vector math.

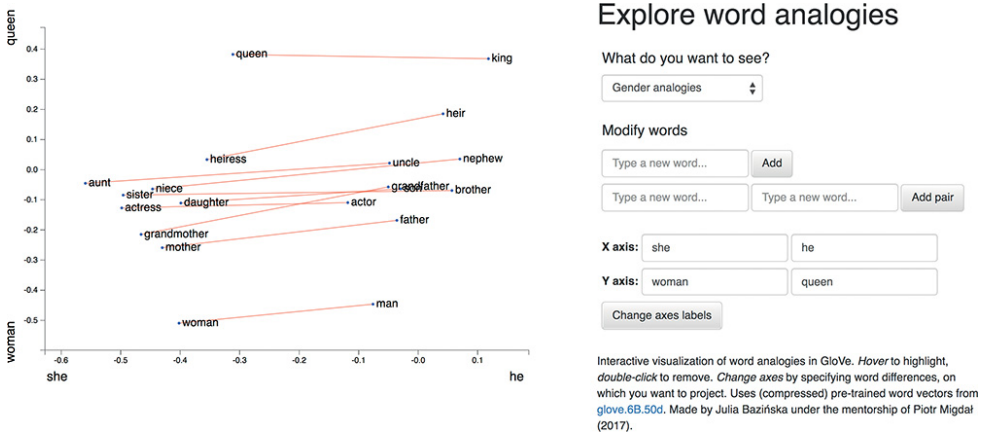


(a)

You remember vectors, right?

$$v_{king} - v_{man} + v_{woman} = v_{queen}$$
$$v_{bezos} - v_{amazon} + v_{tesla} = v_{musk}$$
$$v_{windows} - v_{microsoft} + v_{google} = v_{android}$$

Illustrative word vector arithmetic



Screenshot from Word2viz

Sources: PressBooks, College Physics: OpenStax, Chapter 17 Figure 9, and Krohn (2019), *Deep Learning Illustrated*, Figures 2-7 & 2-8.

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Pretrained word embeddings

GloVe (_Gl_obal _Ve_ctors) are pre-trained word embeddings from Stanford, trained on Wikipedia + Gigaword (6 billion tokens, ~400,000 word vocabulary).

```
1 f"The size of the vocabulary is {len(words)}"
```

```
'The size of the vocabulary is 400000'
```

Look up a word vector

```
1 vec("pizza")
```

```
array([ 0.04,  0.07,  0.06, -0.  , -0.03,  0.03, -0.01, -0.04, -0.07,
        0.01, -0.04, -0.1 , -0.03,  0.09, -0.05,  0.  , -0.02, -0.02,
       -0.05,  0.06,  0.05,  0.11, -0.01, -0.02,  0.06, -0.01,  0.04,
        0.  ,  0.06, -0.18, -0.08,  0.07,  0.05, -0.04, -0.08,  0.05,
       -0.06, -0.05, -0.07,  0.04,  0.02,  0.01,  0.  ,  0.08, -0.02,
        0.05,  0.15,  0.02,  0.01, -0.03, -0.01, -0.1 ,  0.05,  0.08,
       -0.03, -0.04, -0.01,  0.05,  0.02, -0.04,  0.12, -0.04, -0.  ,
       -0.07, -0.02, -0.05, -0.03,  0.07, -0.04,  0.04,  0.08,  0.02,
       -0.01, -0.06,  0.02, -0.03, -0.02, -0.01, -0.01, -0.05, -0.02,
        0.06,  0.01, -0.06, -0.02, -0.07,  0.04,  0.04, -0.05,  0.01,
        0.04,  0.02, -0.02, -0.02, -0.  ,  0.01, -0.04,  0.03, -0.06,
        0.01,  0.05, -0.01,  0.  , -0.07, -0.01, -0.06,  0.02,  0.11,
       -0.03,  0.02,  0.05,  0.  ,  0.04, -0.09,  0.06, -0.09, -0.09,
        0.04, -0.05, -0.01, -0.03,  0.02,  0.12, -0.02, -0.04,  0.03,
        0.01,  0.02, -0.05, -0.02,  0.01,  0.06, -0.03, -0.  ,  0.03,
       -0.03,  0.  ,  0.03, -0.02,  0.06,  0.02,  0.01, -0.04, -0.06,
```

```
1 len(vec("pizza"))
```

```
300
```

UNSW SYDNEY

Find nearby word vectors

```
1 nearest(vec("python"))
```

```
[('python', 1.0),
 ('monty', 0.683738112449646),
 ('perl', 0.519283652305603),
 ('cleese', 0.5092198848724365),
 ('pythons', 0.5007114410400391),
 ('php', 0.4942314028739929),
 ('grail', 0.4683017134666443),
 ('scripting', 0.467612624168396),
 ('skit', 0.4474538564682007),
 ('javascript', 0.4312553405761719)]
```

```
1 similarity("python", "java")
```

```
0.35804617404937744
```

```
1 similarity("python", "sport")
```

```
0.005185101181268692
```

```
1 similarity("python", "r")
```

```
0.060789704322814944
```

What does ‘similarity’ mean?

The ‘similarity’ scores

```
1 similarity("sydney", "melbourne")
```

```
0.7951619625091553
```

are normally based on cosine distance.

```
1 x = vec("sydney")
2 y = vec("melbourne")
3 x.dot(y) / (np.linalg.norm(x) * np.linalg.norm(y))
```

```
np.float32(0.795162)
```

```
1 similarity("sydney", "aarhus")
```

```
0.14399726688861847
```

Weng's GoT Word2Vec

In the Game of Thrones (GoT) word embedding space, the top similar words to "king" and "queen" are:

```
model.most_similar("king")
```

```
('kings', 0.897245)
('baratheon', 0.809675)
('son', 0.763614)
('robert', 0.708522)
('lords', 0.698684)
('joffrey', 0.696455)
('prince', 0.695699)
('brother', 0.685239)
('aerys', 0.684527)
('stannis', 0.682932)
```

```
model.most_similar("queen")
```

```
('cersei', 0.942618)
('joffrey', 0.933756)
('margaery', 0.931099)
('sister', 0.928902)
('prince', 0.927364)
('uncle', 0.922507)
('varys', 0.918421)
('ned', 0.917492)
('melisandre', 0.915403)
('robb', 0.915272)
```

Source: Lilian Weng (2017), Learning Word Embedding, Blog post.

Combining word vectors

You can summarise a sentence by averaging the individual word vectors.

```
1 sv = (vec("melbourne") + vec("has") + vec("better") + vec("coffee")) / 4
2 len(sv), sv[:5]
```

```
(300, array([-0.03,  0.03,  0.01,  0.01, -0.01], dtype=float32))
```

> "As it turns out, averaging word embeddings is a surprisingly effective way to create word embeddings. It's not perfect (as you'll see), but it does a strong job of capturing what you might perceive to be complex relationships between words."
>
> — Trask (2019, ch. 12)

Analogies with word vectors

```
1 analogy("paris", "france", "london")
```

```
[('britain', 0.7673852443695068),
 ('england', 0.6279442310333252),
 ('uk', 0.6197735667228699),
 ('british', 0.6013829708099365),
 ('ireland', 0.5365166664123535),
 ('u.k.', 0.5294836759567261),
 ('scotland', 0.5195812582969666),
 ('australia', 0.5150107145309448),
 ('wales', 0.5144591927528381),
 ('europe', 0.5047693848609924)]
```

```
1 analogy("man", "king", "woman")
```

```
[('queen', 0.6713277101516724),
 ('princess', 0.5432624816894531),
 ('throne', 0.538610577583313),
 ('monarch', 0.5347575545310974),
 ('daughter', 0.4980251491069737),
 ('mother', 0.49564436078071594),
 ('elizabeth', 0.4832652807235718),
 ('kingdom', 0.47747093439102173),
 ('prince', 0.4668240249156952),
 ('wife', 0.4647327661514282)]
```

Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings

Tolga Bolukbasi[1], **Kai-Wei Chang**[2], **James Zou**[2], **Venkatesh Saligrama**[1,2], **Adam Kalai**[2]
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Extreme *she*	**Extreme *he***
1. homemaker	1. maestro
2. nurse	2. skipper
3. receptionist	3. protege
4. librarian	4. philosopher
5. socialite	5. captain
6. hairdresser	6. architect
7. nanny	7. financier
8. bookkeeper	8. warrior
9. stylist	9. broadcaster
10. housekeeper	10. magician

Gender stereotype *she-he* analogies

sewing-carpentry	registered nurse-physician	housewife-shopkeeper
nurse-surgeon	interior designer-architect	softball-baseball
blond-burly	feminism-conservatism	cosmetics-pharmaceuticals
giggle-chuckle	vocalist-guitarist	petite-lanky
sassy-snappy	diva-superstar	charming-affable
volleyball-football	cupcakes-pizzas	lovely-brilliant

Gender appropriate *she-he* analogies

queen-king	sister-brother	mother-father
waitress-waiter	ovarian cancer-prostate cancer	convent-monastery

Source: The title block and Figure 1 directly from Bolukbasi et al. (2016).

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The vectorisation toolkit

Every method turns text into numbers; they differ on *what* becomes a vector — a single word or a whole document — and *how* that vector looks (sparse vs dense):

	One *word* → vector	**One *document* → vector**
Sparse length |V|, one column per word	one-hot encoding	bag of words, TF-IDF
Dense short, learned dimensions	word embeddings	averaged word embeddings

- *Sparse* vectors have one *interpretable* column per vocabulary word; *dense* embeddings are short and their columns are uninterpretable learned dimensions.
- A *document* vector is just its word vectors *pooled*: sum the one-hots for bag of words; average the embeddings for a sentence vector.
- Because embeddings are word-level, representing a whole document needs an extra step — averaging them, or feeding the sequence into an RNN.

Package Versions

```
1 from watermark import watermark
2 print(watermark(python=True, packages="keras,matplotlib,numpy,pandas,seaborn,scipy,torch"
```

```
Python implementation: CPython
Python version       : 3.14.5
IPython version      : 9.15.0

keras     : 3.15.0
matplotlib: 3.11.0
numpy     : 2.5.0
pandas    : 3.0.3
seaborn   : 0.13.2
scipy     : 1.18.0
torch     : 2.12.1
```

Glossary

- bag of words
- lemmatization
- n-grams
- one-hot embedding
- TF-IDF
- vocabulary
- word embedding
- word2vec

References

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